Thomas VAUDESCAL

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Personal profile

Quantitative Researcher at Synerqo, focused on **systematic options trading, vol surface modeling, market microstructure, and systematic quoting**. Hands-on across the full research-to-production loop: in-house options backtesting engine (Python/Numba), pricing engine for exotics and structured products with Greeks via AAD, low-latency Rust execution components, and live PnL ownership on a GEX-driven SPX options strategy. MSc in Financial Engineering, HEC Montréal. Prior: Quantitative Analyst at Deloitte (credit risk modeling), Quantitative Developer at National Bank Financial Markets (SmartBeta strategies, $800M AUM). Thesis: **Gamma Exposure and Intraday Volatility Dynamics in the SPX Options Market** – dealer-hedging-induced regime shifts in intraday vol surface dynamics (5-minute SPX options data, 2B points, August 2020–July 2025).

Professional Experience

Synerqo

Montreal, Canada

Quantitative Researcher — *May 2025 - Present*

- Designed, backtested (QuantConnect + in-house engine) and **deployed live a systematic options strategy driven by dealer gamma exposure (GEX)** in the SPX market. Signal directly derived from my MSc thesis research on dealer hedging and intraday volatility dynamics.
- Researched and prototyped a **systematic options market-making strategy**: skew-aware quoting, inventory-driven price skewing, and adverse-selection filters built on order-book microstructure features.
- Built **low-latency execution components in Rust** (market-data ingestion, hot-path quote updates, order routing) with end-to-end latency profiling against the Python research stack.
- Built and systematized additional options strategies (Volatility Risk Premium, systematic put-writing), reaching **25% annualized returns** over a 7-year backtest with a controlled drawdown ($\sim$25%).
- Built **from scratch an event-driven options backtesting engine in Python + Numba**: vectorized hot paths, realistic friction modeling (bid-ask, slippage, commissions, margin, assignment), used alongside QuantConnect to validate every derivatives strategy before live deployment.
- Built **from scratch a pricing engine for exotic options and structured products**, with Greeks computed via **Adjoint Algorithmic Differentiation (AAD)**. Monte Carlo pricing, PDE solvers, automated delta/gamma/vega hedging, and calibration of stochastic volatility models (Heston, SABR, SVI) robust to surface dislocations (term-structure inversions, skew regime shifts).
- Developed complementary systematic strategies on US equities, indices, fixed income (Momentum, Mean Reversion) and **intraday FX** (microstructure-driven signals on liquid majors), all running on a Python cloud-native execution and monitoring stack with 7-figure monthly notional.
- **Stack**: Python (Numba, NumPy, asyncio), C/C++, AAD, QuantLib, cloud infrastructure, custom backtesting & pricing engines, risk analytics.

Financial Engineering & Modeling - Financial Risk | Deloitte

Montreal, Canada

Quantitative Analyst — *October 2024 - June 2025*

- Developed and validated credit risk models (PD, LGD, EAD) for major Canadian financial institutions under IFRS 9 and Basel III/IV regulatory frameworks.
- Led independent model validation engagements: statistical backtesting, sensitivity analysis, stress testing, and regulatory documentation for OSFI compliance.
- Performed quantitative assessment of model assumptions, calibration methodologies, and discriminatory power using Gini coefficients and KS statistics.
- Built automated model monitoring dashboards and validation pipelines in Python, reducing manual review time by **40%**.
- Authored technical validation reports and presented findings to senior stakeholders and client risk committees.
- **Stack**: Python, R, SQL, statistical modeling, regulatory compliance (IFRS 9, Basel III/IV, OSFI guidelines).

Global Equity Derivatives R&D - Structured Products & Exotics | National Bank Financial Markets

Montreal, Canada

Quantitative Developer Intern — *May 2023 - December 2023*

- Developed and deployed SmartBeta strategies (Low Volatility, Momentum) managing **$800M AUM** within a $7B structured products division.
- Achieved **$1M annual cost savings** through strategy optimization and migration from external vendor to internal production systems.
- Built end-to-end backtesting framework in Python: signal generation, portfolio construction, transaction cost modeling, risk analytics, and performance attribution.
- Implemented portfolio optimization algorithms with constraints (turnover limits, sector exposure, factor tilts) using convex optimization (cvxpy).
- Delivered weekly quantitative reports to senior stakeholders and presented strategy performance and risk metrics to trading desk and portfolio managers.
- Collaborated with structuring and trading teams to integrate quantitative signals into exotic derivatives pricing and hedging workflows.
- **Stack**: Python (OOP), Bloomberg Terminal & API, SQL, AWS (EC2, S3), optimization algorithms, statistical testing.

Videns Analytics

Montreal, Canada

Data Analyst — *April 2020 - June 2021*

- Designed and delivered data science curriculum (Python, statistics, ML) for *Videns Academy* platform. Content adopted by UQAC for a university course.
- Built predictive models and automated reporting pipelines for insurance clients, improving claims analysis and risk assessment workflows.
- Led consulting engagements in insurance analytics: data cleaning, exploratory analysis, and dashboard development for client decision-making.
- Promoted from intern to full-time Data Analyst based on project delivery and client satisfaction.

Education

HEC Montreal
Montreal, Canada

MSc in Financial Engineering - Thesis *August 2022 - December 2024*

- **Thesis:** Gamma Exposure and Intraday Volatility Dynamics in the SPX Options Market. Mapping of dealer-hedging-induced regime shifts in intraday vol surface dynamics using 5-minute options data (2B+ data points, Aug 2020–Jul 2025).
- **GPA:** 3.88/4.3. Recipient of the **Deloitte M.Sc.**, **Alpha Fixe Capital**, **Jocelyne & Jean C. Monty I**, and **Jocelyne & Jean C. Monty II** scholarships of excellence for academic excellence.
- **Hackathons:** ConUHacks VIII winning team for National Bank challenge 2024 (1000+ participants). Top 10 finalist at McGill FIAM Portfolio Management Hackathon (66 teams). 2nd place at the HEC Montreal Trading Club Trading Competition.
- **Leadership:** Vice President Research at the Trading Club, organized workshops on derivatives and systematic trading. R&D Analyst at BNI-HEC Investment Fund ($5M AUM), conducted equity research and portfolio analysis.
- **Courses:** Stochastic Calculus, Derivatives Pricing, Fixed Income Securities, Financial Econometrics, Machine Learning for Finance, Numerical Methods, Risk Management.

HEC Montreal
Montreal, Canada

Bachelor's degree in Business Administration | specialization in Mathematics, Finance and Economics *August 2017 - May 2022*

- **GPA:** 3.8/4.3 for specialization courses. Selective quantitative track combining advanced mathematics, economics, and finance.
- **Leadership:** Vice President Education at the Data Science Committee, organized Python workshops and machine learning seminars.
- **Competitions:** RITC Trading Competition participant, simulated trading in equities and derivatives markets.
- **Teaching:** Teaching Assistant for Intermediate Microeconomics (200+ students). Private mathematics tutor (Calculus, Linear Algebra).
- **Courses:** Econometrics, Portfolio Management, Options & Futures, Quantitative Finance, Probabilistic Models, Corporate Finance.

University of Montreal
Montreal, Canada

Minor in Mathematics *August 2018 - August 2022*

- Complementary mathematics program completed concurrently with BBA to strengthen quantitative foundations.
- **Courses:** Probability Theory, Mathematical Statistics, Stochastic Processes, Linear Algebra, Multivariable Calculus, Real Analysis, Programming (C/C++).

Skills

Programming	**Python** (5+ years): NumPy, Numba, Pandas, SciPy, statsmodels, scikit-learn, PyTorch, cvxpy, QuantLib, asyncio. **C++** (modern, low-allocation hot paths). **Rust**: tokio, serde, axum. **SQL**, R, Matlab, VBA.
Web	TypeScript, React, Next.js, Node.js, Tailwind CSS, REST & WebSocket APIs.
Quant Finance	Options pricing (vanilla, exotic, structured products), Greeks via AAD, stochastic volatility calibration (Heston, SABR), gamma exposure analysis, backtesting engine design, portfolio optimization, risk management, market microstructure, systematic trading, Monte Carlo simulation.
Data & Tools	Bloomberg Terminal, large-scale data processing (2B+ records), Git, Docker, AWS, REST APIs, WebSocket (live data), Linux/Shell.
Research	Scientific writing, quantitative research methodology, statistical analysis, academic publications, technical presentations.

Languages

French	Native
English	Professional proficiency (spoken and written)

Interests

Sports	Marathon running, mountain hiking, indoor climbing, swimming, weight training, scuba diving.
Intellectual	Chess (1900 Elo), mathematics, philosophy, programming.